INCH-POUND

MIL-DTL-85097/13D(OS)
9 June 2026
SUPERSEDING
MIL-DTL-85097/13C(OS)
20 April 2000

DETAIL SPECIFICATION SHEET

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS) ROCKET MOTOR, CANOPY JETTISON, MARK 109 MODS 0 AND 1 (LOADED ASSEMBLIES)

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-85097.

1. SCOPE

1.1 Scope. This specification covers the Aircrew Escape Propulsion Systems (AEPS) Canopy Jettison Rocket Motor, Mark 109 Mods 0 and 1 (Loaded Assemblies) (see 6.1).

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-P-83126 - Propulsion Systems, Aircrew Escape, Design Specification for

MIL-DTL-85097 - USN/USMC Propellant Actuated Devices (PADs), General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-464 - Electromagnetic Environmental Effects Requirements for Systems

(Copies of these publications are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05T, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

AMSC N/A FSC 1377

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:24Z
Check the source to verify that this is the current version before use.

NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWINGS

1507AS100 - Rocket Motor, Canopy Jettison, Mk 109 Mod 0

1507AS132 - Tube, Motor

1507AS201 - Rocket Motor, Canopy Jettison, Mk 109 Mod 1

(Copies of these publications are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Data Management Branch (Code E26), 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092. Refer questions, inquiries, or problems to nswc_iheodtd_dmo@navy.mil.)

NAVAL SURFACE WARFARE CENTER INDIAN HEAD DIVISION (NSWC IHD) PUBLICATIONS

510-174TD-004 - Propellant, NSWC/IH-EC-8

511-174TD-004 - Propellant, NOSIH-BC-11, NSWC/IH-EC-02, NSWC/IH-EC-04, and NSWC/IH-EC-05

DTL-WS-35787 - Propellant, NSWC/IH-EC-8

(Copies of these publications are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Data Management Branch (Code E26), 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092. Refer questions, inquires, or problems to nswc_iheodtd_dmo@navy.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 General. The requirements for acquiring the product described herein shall consist of this document and MIL-DTL-85097.

3.2.1 Design. The rocket motors described herein shall conform to 1507AS100 (Mk 109 Mod 0) or 1507AS201 (Mk 109 Mod 1).

3.3 Propellant aging. When tested in accordance with MIL-DTL-85097, the propellant shall meet the aging requirements of 511-174TD-004 (Mk 109 Mod 0) or DTL-WS-35787 (Mk 109 Mod 1).

3.4 Propellant. The propellant shall conform to the requirements of 511-174TD-004 (Mk 109 Mod 0) or DTL-WS-35787 (Mk 109 Mod 1).

3.5 Ballistic performance requirements. The rocket motors shall meet the ballistic performance requirements of table I when test fired at both -65±5 °F and 165±5 °F.

TABLE I. Ballistic performance requirements.

Parameter	-65±5 °F		+165±5 °F	
	Min.	Max.	Min.	Max.
Total impulse (lb-sec)	178	211	181	216
Action time (ms)	125	172	88	143
Max thrust (lb)	1305	1933	1665	2746
Thrust angle (deg)	86	94	86	94
Time to 70% max thrust (ms)	-	20	-	20

3.6 Hydrostatic proof requirement. Each motor tube shall meet the hydrostatic test requirements of Drawing 1507AS132.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:24Z
Check the source to verify that this is the current version before use.

3.7 Leakage. Each rocket motor assembly shall have a leakage rate of less than 1×10^{-4} cc/sec air when subjected to the dry leak test of 4.4.5.

3.8 Threaded joint torque. The 1.890-24NS-3 threaded joints (2) of each rocket motor assembly shall withstand 400-inch-pounds torque without breakaway, when tested in accordance with MIL-P-83126.

3.9 Electrical bonding resistance. The electrical bonding resistance between the shielded mild detonating cord (SMDC) housing and the mounting lugs of each rocket motor assembly shall be no greater than 0.0025 ohm, when tested in accordance with MIL-STD-464.

3.10 Motor weight. The weight of the rocket motor shall not exceed 3.50 pounds.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with table I of MIL-DTL-85097.

4.3 Conformance inspection. Conformance tests shall be performed as specified in MIL-DTL-85097 as described in 4.4 through 4.4.7.

4.4 Examinations and tests.

4.4.1 Propellant test results. Propellant test results shall be submitted in accordance with MIL-DTL-85097 and 511-174TD-004 (Mk 109 Mod 0) or MIL-DTL-85097 and DTL-WS-35787 (Mk 109 Mod 1).

4.4.2 Ballistic performance tests. The rocket motor shall be fired using a blasting cap and detonation cord test tip (SMDC line) to fire the primer, igniting the ignition charge and rocket motor grain. The test shall satisfy ballistic parameters specified in 3.5. The sample conditioning temperatures and the time from the conditioning chamber to test firing shall be in accordance with 4.4.2.1 and 4.4.2.2. Failure of any unit to meet ballistic requirements specified in 3.5 shall be cause for rejection of the lot.

4.4.2.1 Conditioning temperature. The conditioning temperatures and tolerances shall be as follows:

a. Cold: -65±5 °F

b. Hot: 165±5 °F

4.4.2.2 Conditioning period. Time for temperature conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: 4 hours, minimum

b. Temperature cycling: 5 hours, minimum

4.4.3 Radiographic inspection. All components or assemblies shall be radiographed in two views 90 degrees apart.

4.4.4 Hydrostatic proof test. Each motor tube shall be subjected to the hydrostatic tests of Drawing 1507AS132.

4.4.5 Leakage test. Each rocket motor assembly shall be subjected to a dry leak test to verify compliance with a leakage rate of less than 1×10^{-4} cc/sec air.

4.4.6 Threaded joint torque test. The 1.890-24NS-3 threaded joints (2) of each rocket motor assembly shall be subjected to the torque test of MIL-P-83126.

4.4.7 Electrical bonding resistance test. The electrical bonding resistance between the SMDC housing and the mounting lugs of each rocket motor assembly, selected as a sampling test specimen, shall be tested in accordance with MIL-STD-464.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:24Z
Check the source to verify that this is the current version before use.

5. PACKAGING

(This section is not applicable to this specification.)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The rocket motors described herein are intended to be used to eject the canopy during an emergency crew ejection from the F/A/TF-18 Aircraft.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article is required (see 3.1 and 4.2).

c. Mark and Mod designation of rocket motor.

6.3 Definitions.

6.3.1 Action time. Action time is defined as the elapsed time from 10 percent of the maximum thrust on the initial rise of the thrust-time curve to the corresponding 10 percent of maximum thrust on the declining portion of the curve.

6.3.2 Impulse. Impulse is obtained by integrating the thrust of the rocket motor over the action time.

6.3.3 Maximum thrust. Maximum thrust is defined as the maximum recorded thrust that occurs over the action time.

6.3.4 Thrust angle. Thrust angle is defined as the time averaged angle of the thrust vector as measured in the time interval between 10 percent of maximum thrust on the initial rise of the curve and the corresponding 10 percent of maximum thrust on the final declining portion of the curve on the rocket thrust trace.

6.3.5 Time to 70 percent maximum thrust. Time to 70 percent maximum thrust is defined as the elapsed time from the application of an electric current to the blasting cap (test tip) until 70 percent of the maximum thrust.

6.4 Subject term (key word) listing.

Ballistic performance
Detonating cord
Propellant
Radiographic inspection

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – OS
(Project 1377-2025-013)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:24Z
Check the source to verify that this is the current version before use.